University of Cagliari

Department of Engineering

Master's Degree in "Computer Engineering,
Cybersecurity And Artificial Intelligence"

Advanced Embedded System
Lab 4 - Parallel processing

Advisor:
Prof. Paolo Meloni

Authors:
Lello Molinario

Accademic Year 2025/2026
Cagliari

Contents

Lab 4 – Parallel Processing and SIMD Evaluation

This laboratory activity addresses **Lab 4 – Parallel Processing** and focuses on the analysis of task-level and data-level parallelism on the dual-core ARM Cortex-A9 architecture of the Zybo Z7 board. The experiment is conducted in a bare-metal environment and aims at evaluating both the benefits and limitations of parallel execution and SIMD auto-vectorization on a shared-memory embedded platform.

The core objective of the lab is the design and evaluation of a shared-memory parallel application in which **Core 0 (Master)** and **Core 1 (Worker)** cooperate to compute a vector addition. Two input vectors A and B are allocated in shared DDR memory, and the resulting vector C is computed as

$$C[i] = A[i] + B[i].$$

The computational workload is evenly partitioned between the two cores, with Core 0 processing the first half of the vector and Core 1 processing the second half. Inter-core coordination is achieved through a minimal synchronization mechanism based on shared memory flags (`start`, `done0`, `done1`), without any operating system support.

Execution time is measured using the ARM Cortex-A9 Global Timer, allowing a quantitative comparison between serial and parallel implementations. In addition, the impact of data-level parallelism is investigated by enabling compiler auto-vectorization through ARM NEON SIMD instructions. Overall, the laboratory provides insight into low-level multi-core programming, explicit workload partitioning, shared-memory communication, synchronization, and performance evaluation on embedded systems.

System Debugger Configuration

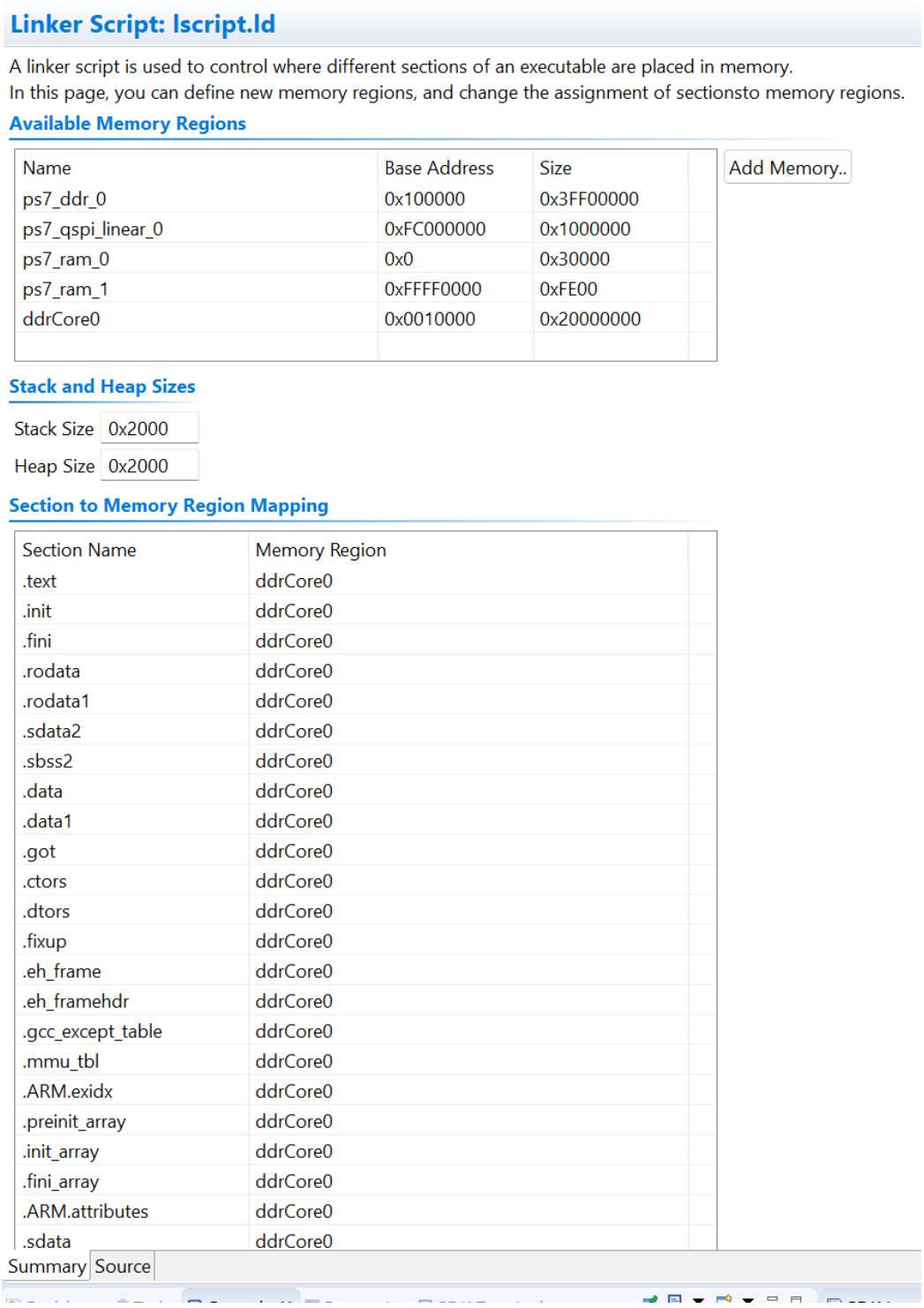
To ensure correct and deterministic execution, a *System Debugger* run configuration was created in the Xilinx SDK environment. Before launching the application, the system is fully reset and the FPGA fabric (PL) is reprogrammed. The Processing System (PS) is initialized through the execution of `ps7_init`, followed by `ps7_post_config`, which enables the level shifters between PL and PS.

The two application binaries are explicitly assigned to the corresponding processors, with `master.elf` mapped to Processor 0 and `worker.elf` mapped to Processor 1. Automatic processor reset and stopping at program entry are disabled to allow both cores to start execution simultaneously. This configuration is essential to obtain reliable timing measurements and correct inter-core synchronization.

Linker Script Modification

In order to support a shared-memory dual-core execution model, the linker script (`lscript.ld`) was explicitly modified for both applications. The memory layout was carefully configured to ensure that executable code, global and static data, stack, and heap are all placed in DDR memory regions accessible by both processors, while avoiding any overlap with the shared communication area.

The shared DDR region used for inter-core communication is excluded from all linker-managed sections, preventing accidental overwriting by either core. This modification is a critical requirement for correctness and determinism in bare-metal multi-core systems and follows the memory layout prescribed in the laboratory instructions.



Linker Script: lscript.ld

A linker script is used to control where different sections of an executable are placed in memory.
In this page, you can define new memory regions, and change the assignment of sectionsto memory regions.

Available Memory Regions

Name	Base Address	Size
ps7_ddr_0	0x100000	0x3FF00000
ps7_qspi_linear_0	0xFC000000	0x1000000
ps7_ram_0	0x0	0x30000
ps7_ram_1	0xFFFF0000	0xFE00
ddrCore0	0x0010000	0x20000000

Add Memory..

Stack and Heap Sizes

Stack Size 0x2000

Heap Size 0x2000

Section to Memory Region Mapping

Section Name	Memory Region
.text	ddrCore0
.init	ddrCore0
.fini	ddrCore0
.rodata	ddrCore0
.rodata1	ddrCore0
.sdata2	ddrCore0
.sbss2	ddrCore0
.data	ddrCore0
.data1	ddrCore0
.got	ddrCore0
.ctors	ddrCore0
.dtors	ddrCore0
.fixup	ddrCore0
.eh_frame	ddrCore0
.eh_framehdr	ddrCore0
.gcc_except_table	ddrCore0
.mmu_tbl	ddrCore0
.ARM.exidx	ddrCore0
.preinit_array	ddrCore0
.init_array	ddrCore0
.fini_array	ddrCore0
.ARM.attributes	ddrCore0
.sdata	ddrCore0

Summary Source

Figure 1: Modified linker script for the Master application (Core 0). All sections are placed in DDR while avoiding overlap with the shared-memory region.

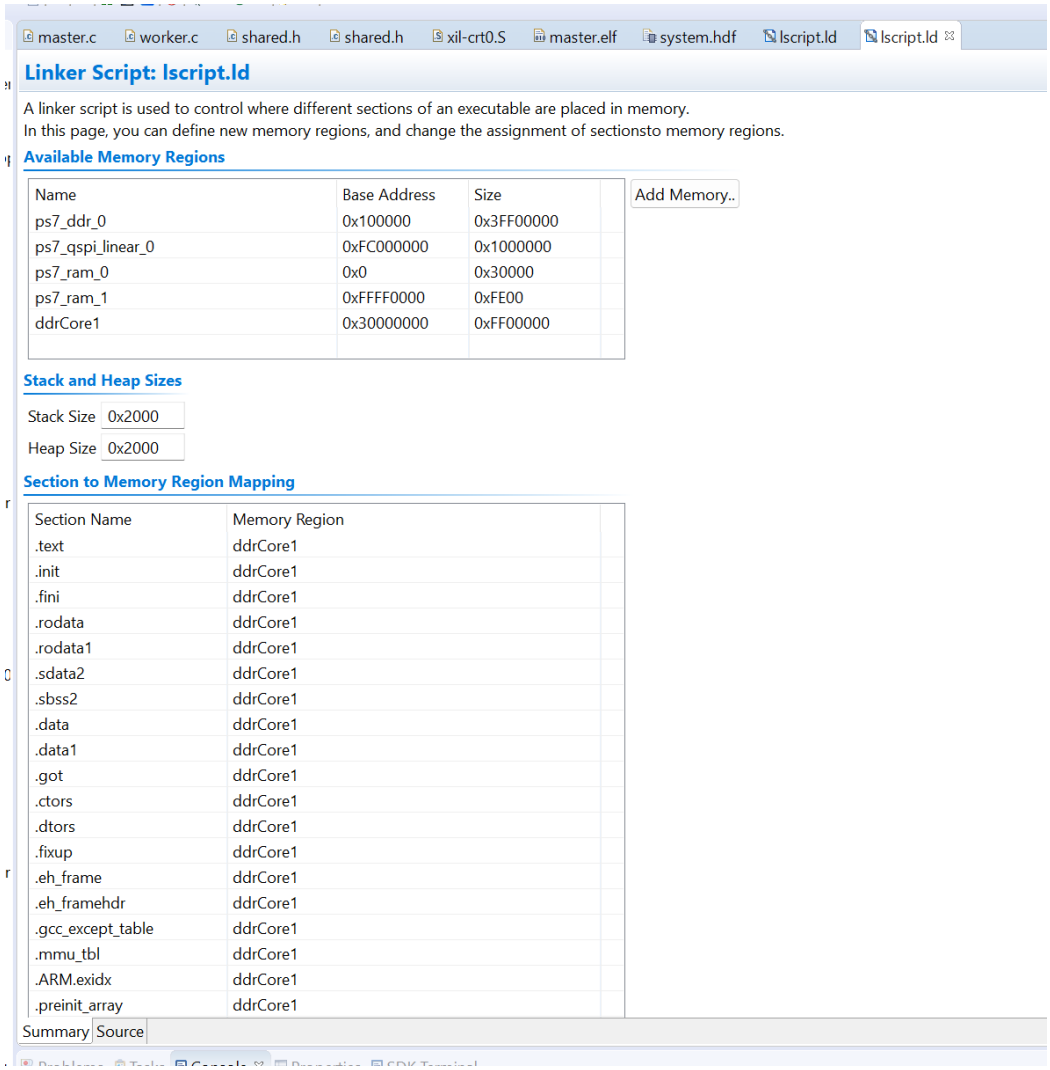


Figure 2: Modified linker script for the Worker application (Core 1). The shared DDR region is excluded from all linker-managed sections.

Serial Baseline Integration

To provide a meaningful reference for performance evaluation, a serial baseline implementation of the vector addition was integrated into the master application. The function executes entirely on Core 0 and serves as a baseline for comparison with the parallel implementation. The use of the `restrict` qualifier enables the compiler to apply aggressive optimizations and auto-vectorization, facilitating the evaluation of SIMD effectiveness independently of task-level parallelism.

Compilation Settings

Both the `master` and `worker` applications were compiled using aggressive optimization settings, explicitly enabling ARM NEON support and compiler auto-vectorization. The selected compiler and linker flags allow the generation of SIMD instructions and provide diagnostic feedback on vectorization decisions. This configuration ensures that both task-level parallelism and instruction-level parallelism are exploited to the maximum extent allowed by the architecture and memory subsystem.

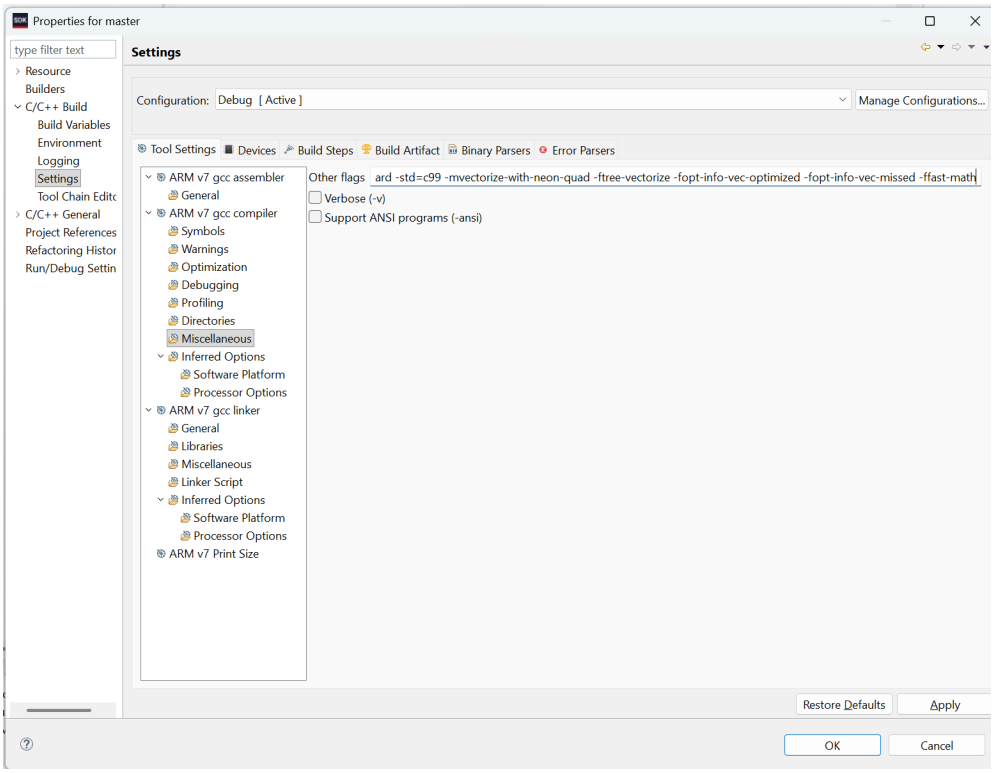


Figure 3: Compiler optimization and auto-vectorization flags used for the Master application.

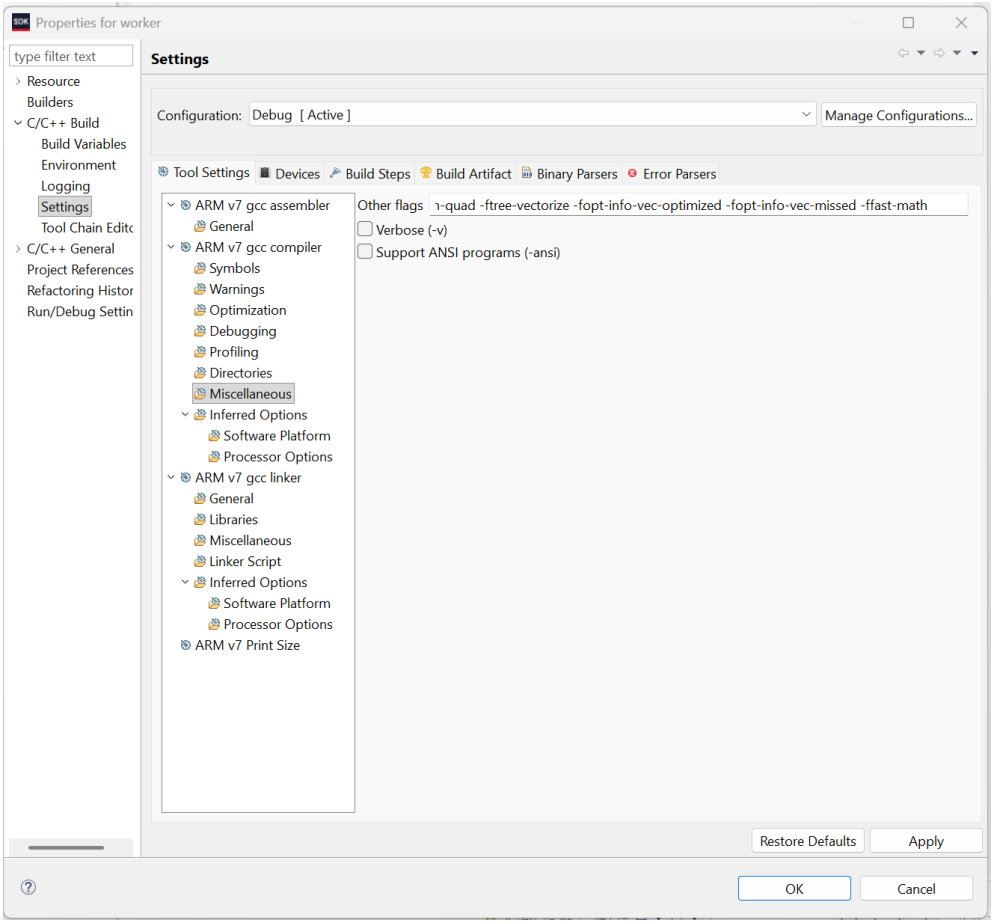


Figure 4: Compiler optimization and auto-vectorization flags used for the Worker application.

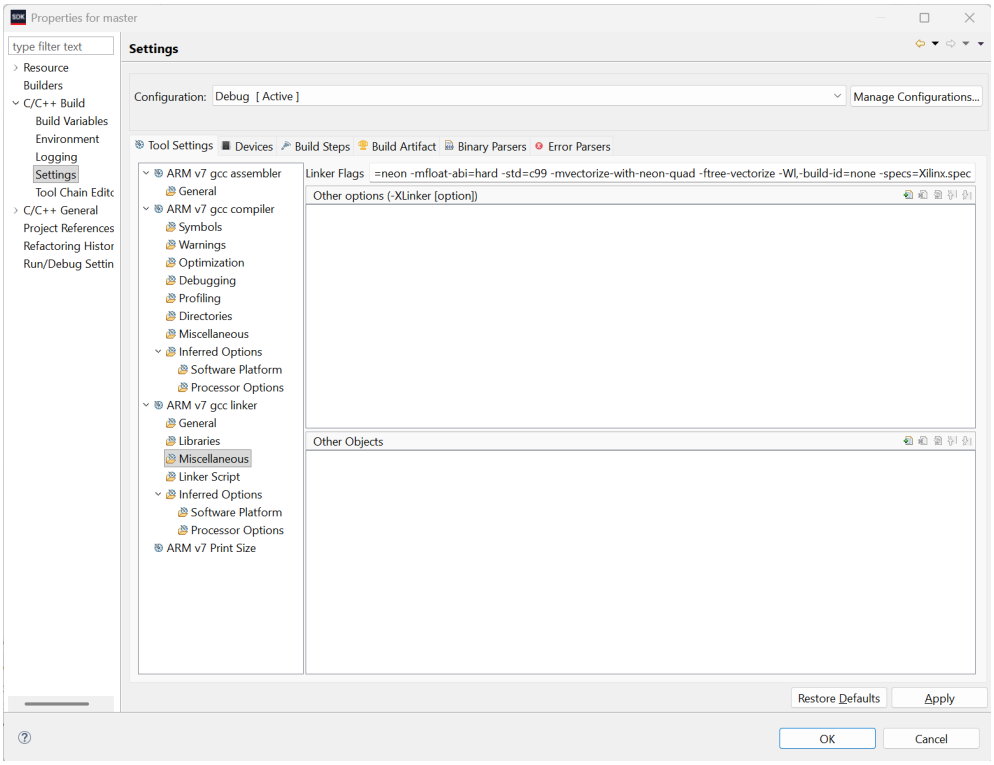


Figure 5: Linker configuration for the Master application.

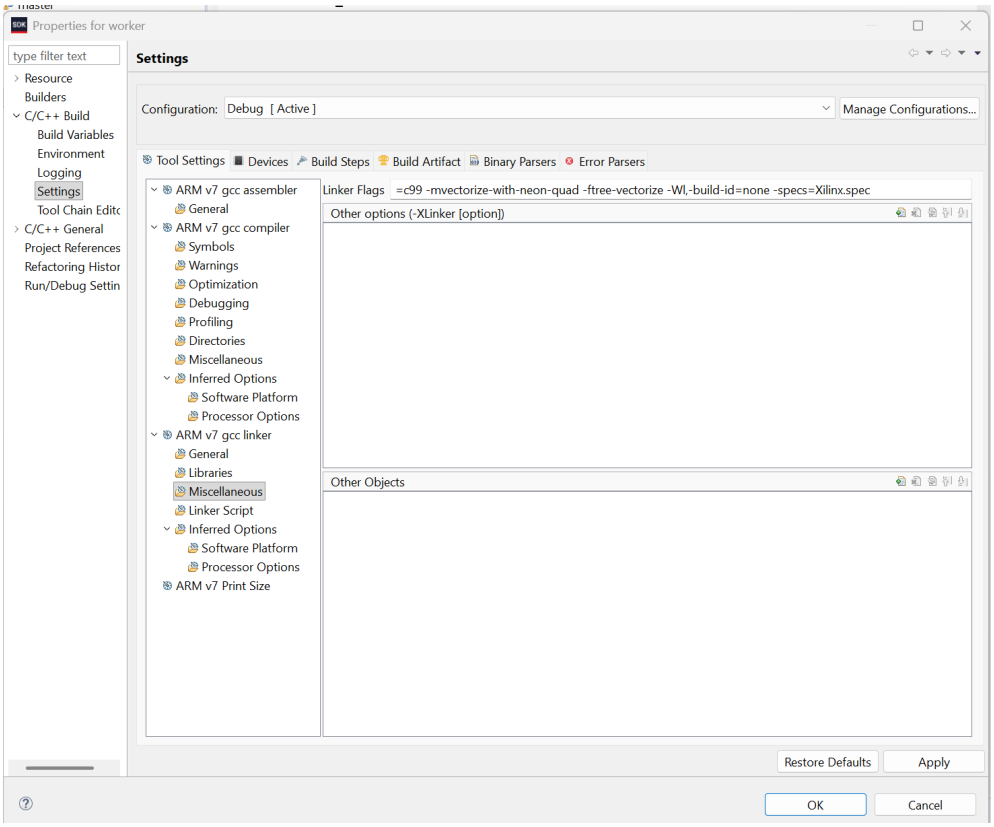


Figure 6: Linker configuration for the Worker application.

Execution Time Measurements

Execution time measurements were performed using the ARM Cortex-A9 Global Timer and expressed in CPU cycles. Experiments were conducted for multiple array sizes (`ARRAY_SIZE = 10`, `100`, `1000`, and `10000`) to assess the scalability of the proposed parallel approach.

The obtained results, summarized in Table 1 and supported by the UART outputs reported in Figures 7–10, show that parallel execution does not provide a significant speedup with respect to the serial baseline. For all tested configurations, the parallel implementation exhibits execution times comparable to, or higher than, the serial version, resulting in a speedup consistently lower than one.

From an architectural perspective, this behavior is expected. The vector addition kernel is strongly memory-bound, and both cores concurrently access the same shared DDR memory. As a consequence, memory bandwidth contention dominates execution time, limiting the benefits of task-level parallelism despite the even workload partitioning and the minimal synchronization overhead.

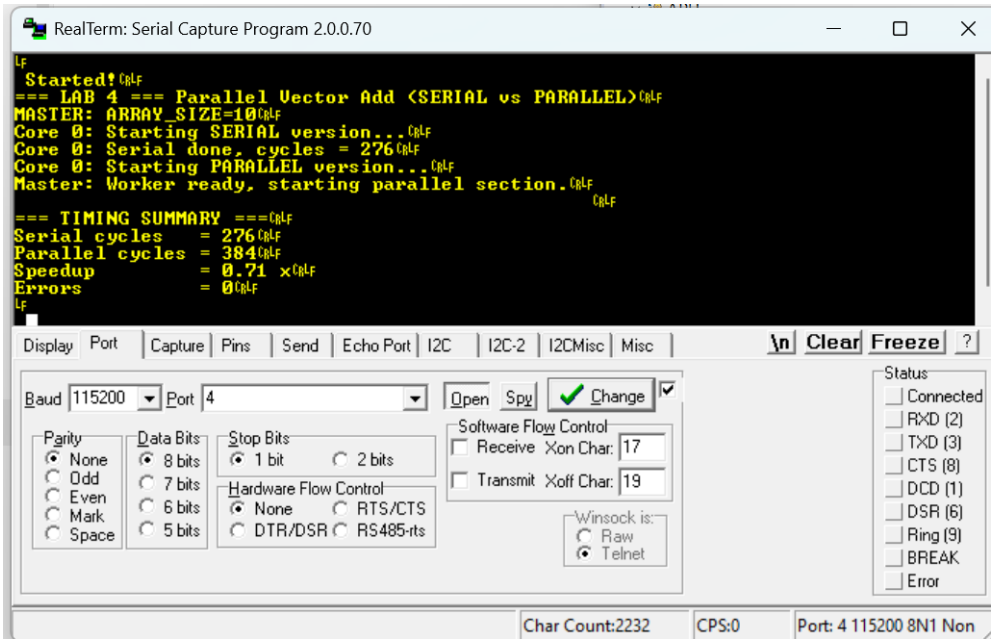


Figure 7: UART output for the parallel vector addition with `ARRAY_SIZE = 10`.

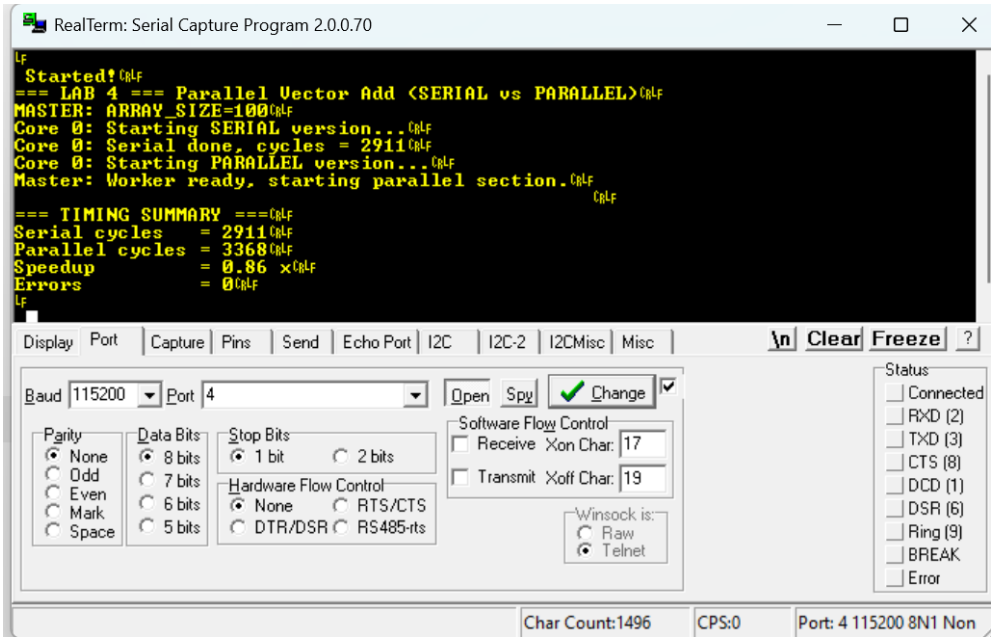


Figure 8: UART output for the parallel vector addition with `ARRAY_SIZE = 100`.

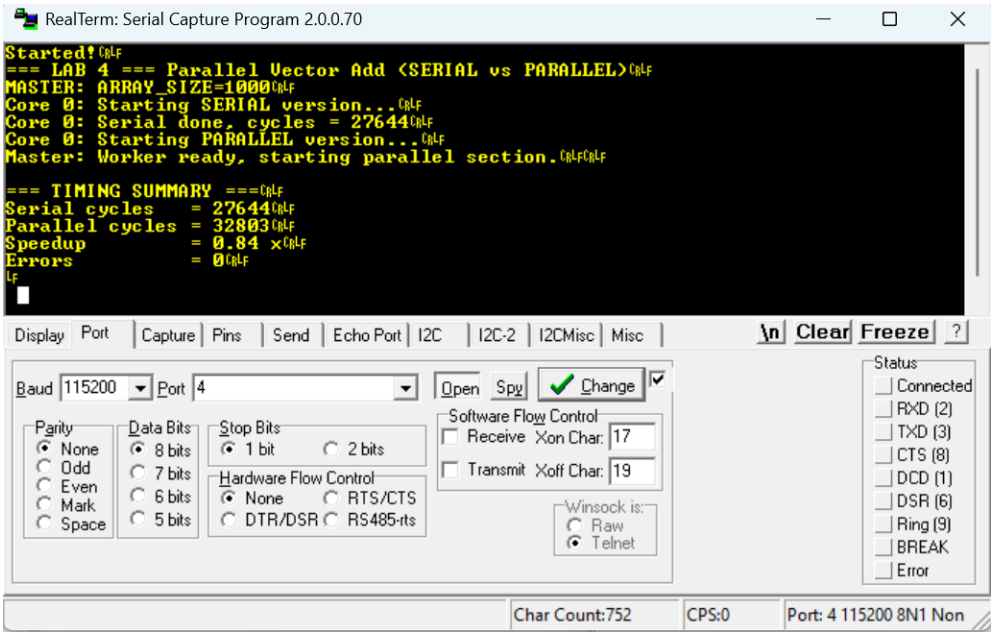


Figure 9: UART output for the parallel vector addition with `ARRAY_SIZE = 1000`.

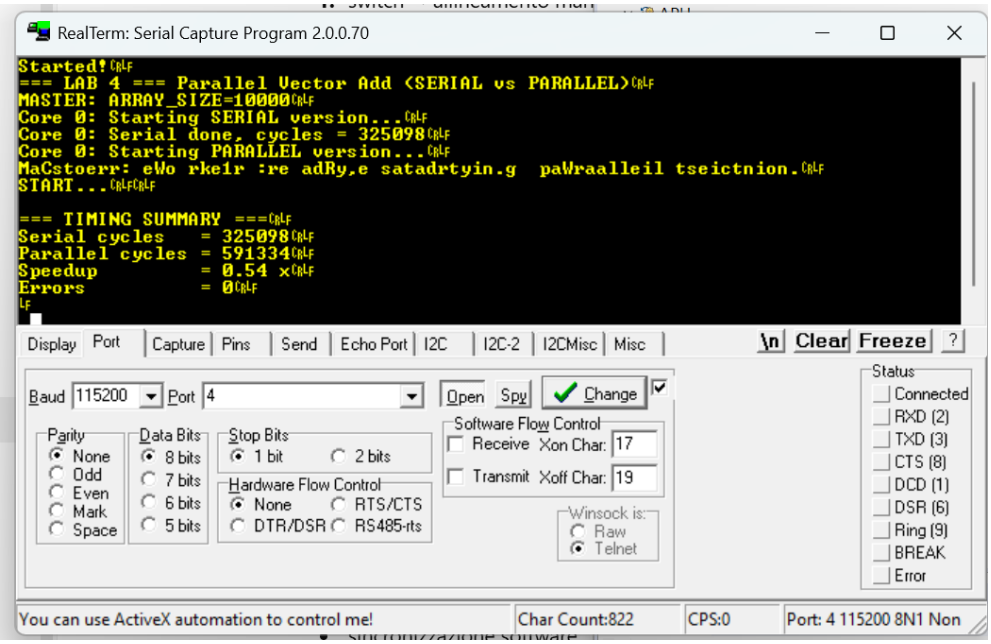


Figure 10: UART output for the parallel vector addition with `ARRAY_SIZE = 10000`.

Table 1: Execution time comparison between serial and parallel implementations for different array sizes (Lab 4).

ARRAY_SIZE	Serial cycles	Parallel cycles	Speedup
10	276	384	0.71×
100	2911	3368	0.86×
1000	27644	32803	0.84×
10000	325098	591334	0.54×

Discussion in Light of Amdahl's Law

The experimental results obtained in Lab 4 can be formally interpreted using *Amdahl's Law*, which provides an upper bound on the achievable speedup of a parallel program as a function of the fraction of code that can be parallelized. According to Amdahl's formulation, the theoretical speedup $S(N)$ obtained using N processing units is given by:

$$S(N) = \frac{1}{(1-P) + \frac{P}{N}},$$

where P represents the fraction of the execution time that can be parallelized, and $(1-P)$ corresponds to the inherently serial portion of the workload.

In the implemented vector addition kernel, the computational loop is fully parallelizable in principle ($P \approx 1$). However, the experimental measurements show speedup values consistently lower than one (Table 1), with measured speedups ranging from approximately 0.71× for `ARRAY_SIZE = 10` to 0.54× for `ARRAY_SIZE = 10000`. This indicates that the effective parallel fraction P is significantly reduced by architectural constraints.

In particular, although the arithmetic operations are parallelized across the two cores, both processors concurrently access the same shared DDR memory. As a result, memory access latency and bandwidth contention dominate execution time, effectively introducing a large serial component that limits scalability. In this context, the memory subsystem behaves as

the true bottleneck of the system, reducing the benefits of task-level parallelism even when synchronization overhead is minimal.

Therefore, the observed lack of speedup is not due to inefficient parallelization, but is a direct consequence of Amdahl's Law applied to a memory-bound workload on a shared-memory embedded architecture. This experimental evidence highlights the importance of considering memory hierarchy and bandwidth constraints when designing parallel applications for embedded multi-core systems.

Assembly Inspection and SIMD Verification

To assess the effectiveness of data-level parallelism, the compiled `master.elf` binary was inspected using the SDK disassembly view. The analysis confirms that the serial baseline loop was successfully auto-vectorized by the compiler.

The presence of ARM NEON instructions, such as vector loads (`vld1.32`), vector additions (`vadd.i32`), and vector stores (`vst1.32`), demonstrates that instruction-level parallelism is effectively exploited. This confirms that the lack of performance improvement in the parallel version is not due to suboptimal code generation, but rather to fundamental memory system limitations.

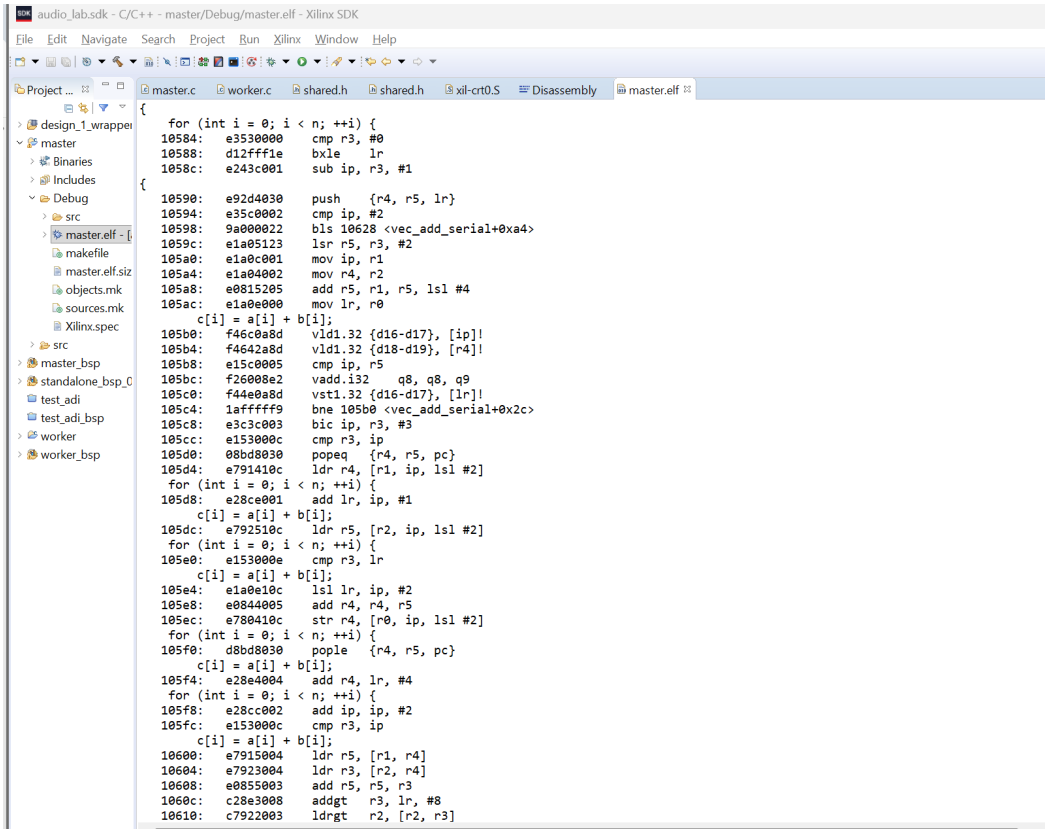


Figure 11: Disassembly of the serial baseline showing ARM NEON SIMD instructions (`vld1.32`, `vadd.i32`, `vst1.32`).

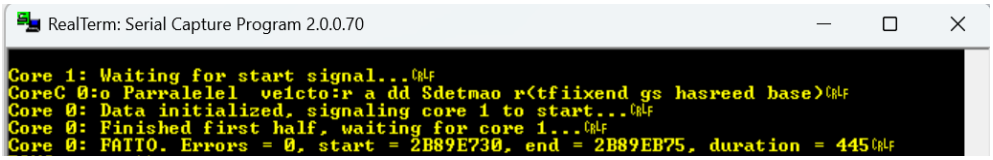


Figure 12: UART output showing correct parallel execution on both cores (Master and Worker).

Results Summary

Execution time measurements were performed using the ARM Cortex-A9 Global Timer to evaluate the impact of parallel execution and SIMD auto-vectorization on a vector addition workload. The following array sizes were considered: `ARRAY_SIZE = 10`, `100`,`1000`, and `10000`.

For all tested configurations, the parallel implementation running on two cores does not provide a significant execution-time improvement compared to the serial baseline. In some cases, the parallel version exhibits similar or slightly higher execution times. This behavior is consistent across all array sizes.

The limited speedup is mainly due to the memory-bound nature of the workload. Both cores concurrently access the same shared DDR memory, resulting in bandwidth contention that dominates execution time and limits the benefits of task-level parallelism.

Assembly inspection confirms that the serial baseline is successfully auto-vectorized by the compiler using ARM NEON SIMD instructions (`vld1.32`, `vadd.i32`, `vst1.32`). While SIMD reduces the number of executed instructions, the overall performance remains constrained by memory access latency rather than computational throughput.

Although both parallel execution and SIMD auto-vectorization were correctly implemented and verified, the experimental results show no execution-time improvement for any of the tested array sizes (`ARRAY_SIZE = 10, 100, 1000, 10000`), due to the memory-bound nature of the workload.

Conclusions

The results of this laboratory are fully aligned with established principles in parallel computing and embedded system design. While both task-level parallelism and SIMD auto-vectorization are correctly implemented and verified, the overall performance is constrained by shared memory bandwidth and access contention.

This lab highlights that parallel execution on multi-core embedded systems does not automatically guarantee performance improvements, particularly for memory-bound workloads. At the same time, it demonstrates the effectiveness of compiler-assisted SIMD vectorization and underscores the importance of memory hierarchy considerations when designing high-performance embedded applications.